



Network Coverage



WPAN

Personal
3–10 m

WLAN

Local
50–300 m

WMAN

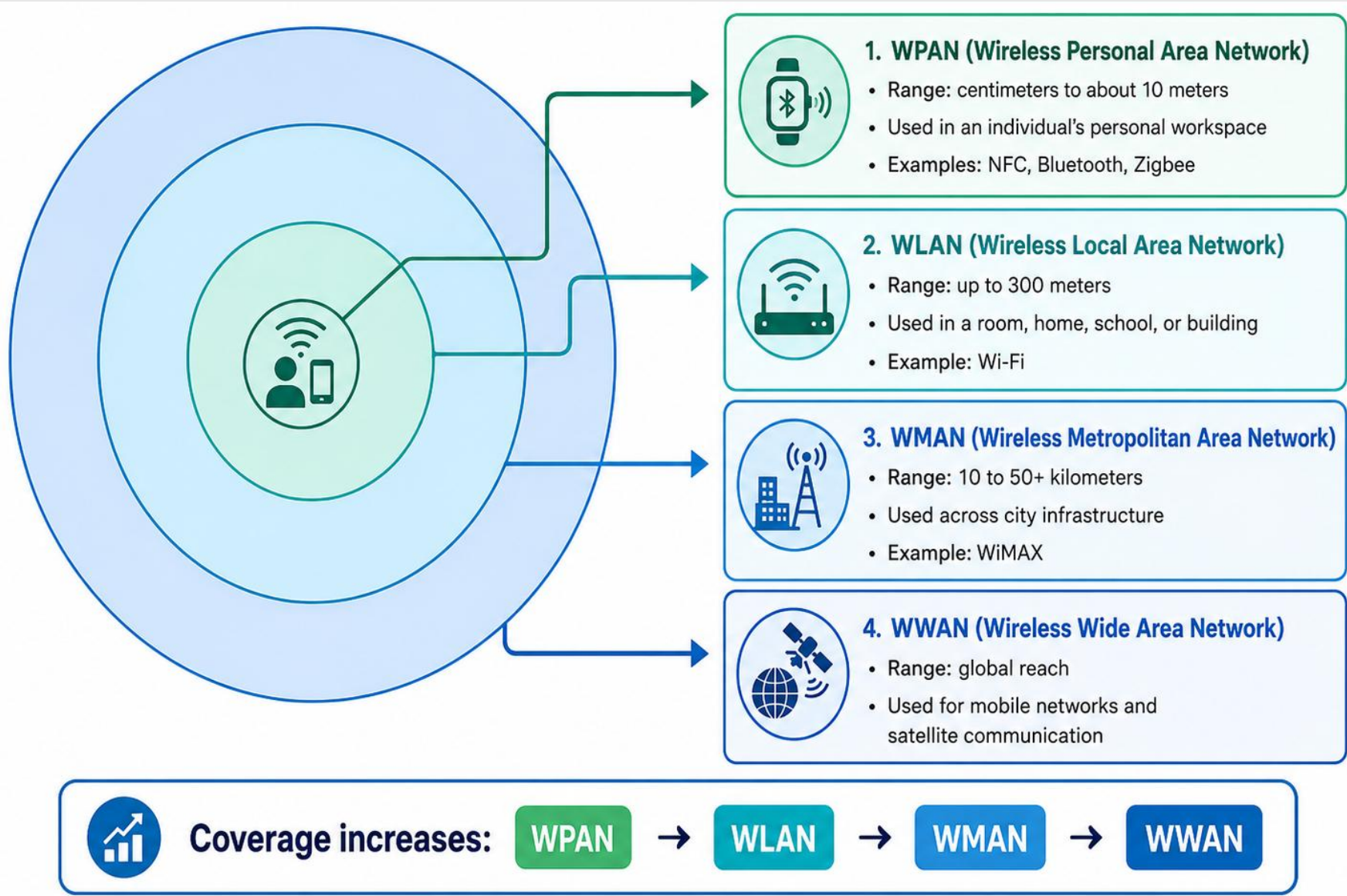
Metropolitan
10–50 km

WWAN

Wide
Country/World



Network Coverage





NFC: Near Field Communication



Full form: Near Field Communication.

Very short-range communication: usually a few centimeters.

Works when two devices are almost touching/tapping.

Used in debit/credit cards, security passes, toll/transport cards and quick information sharing.

NFC = Tap to connect

**Short range gives
better security**

**Example: card payment
or metro pass**



NFC: Near Field Communication



1 CONTACTLESS PAYMENTS

Debit/Credit cards and mobile wallets.



2 ACCESS CONTROL & SECURITY

ID cards, office entry, hotel key cards, security passes.



3 TRANSPORTATION

Bus/train fare cards and toll plazas.



4 DATA SHARING

Sharing contacts, photos, files, or links between devices.



5 SMART TAGS/POSTERS

Tap to get information, websites, or app links.



6 HEALTHCARE/INFORMATION CARDS

Electronic health cards or stored health information.





Zigbee: Low Power IoT Communication



IEEE Standard: 802.15.4.

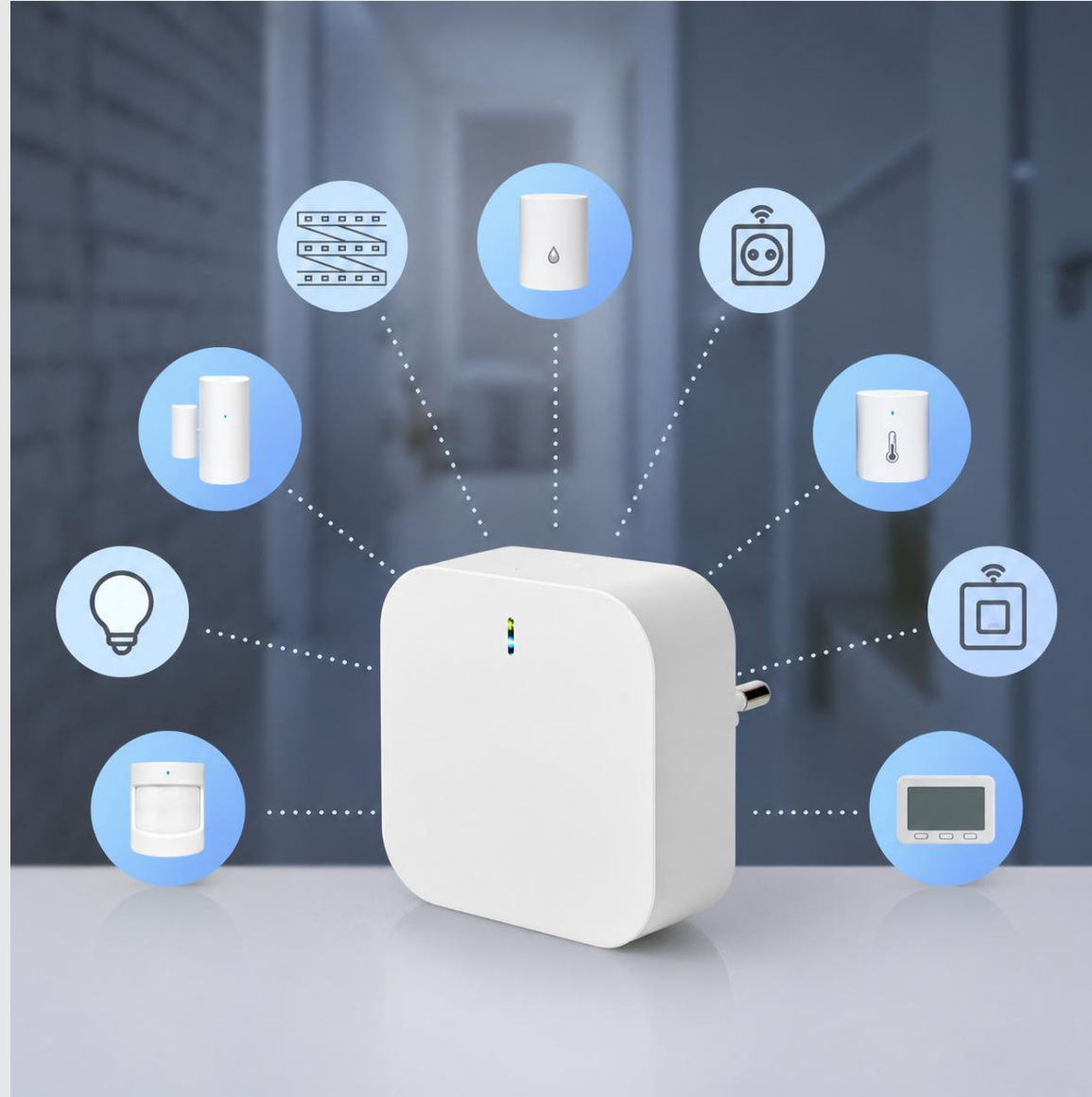
Used in low-powered wireless devices and IoT systems.

Low data rate, low bandwidth and low power consumption.

Examples: smart lights, temperature sensors, smoke detectors, smart meters, industrial automation.



Zigbee: Low Power IoT Communication





Zigbee vs NFC



Feature	Zigbee	NFC
Main idea	Low-power IoT/sensor network	Very short-range tap communication
Range	Short/room-level sensor network	Few centimeters
Use	Smart home, sensors, automation	Payment, ID/pass, transport card
Best phrase	Low power, low bandwidth	Tap and go



Bluetooth: Short-range Personal Connection



Wireless technology for communication between nearby devices.

Works around 2.45 GHz frequency band.

Network type: WPAN / Personal Area Network.

Common uses: earbuds, speakers, mouse, keyboard, smartwatch, file sharing.

IEEE Standard: 802.15

**Typical range:
3–10 meters**

**Low power
Lower speed**

Half Duplex Communication



Bluetooth: Short-range Personal Connection



Technical Dashboard

Standard

IEEE 802.15

Frequency

2.45 GHz

Base Range

3 to 10 meters (up to 100m with power modifications)

Transmission Mode

Half-duplex

Evolution of Speed

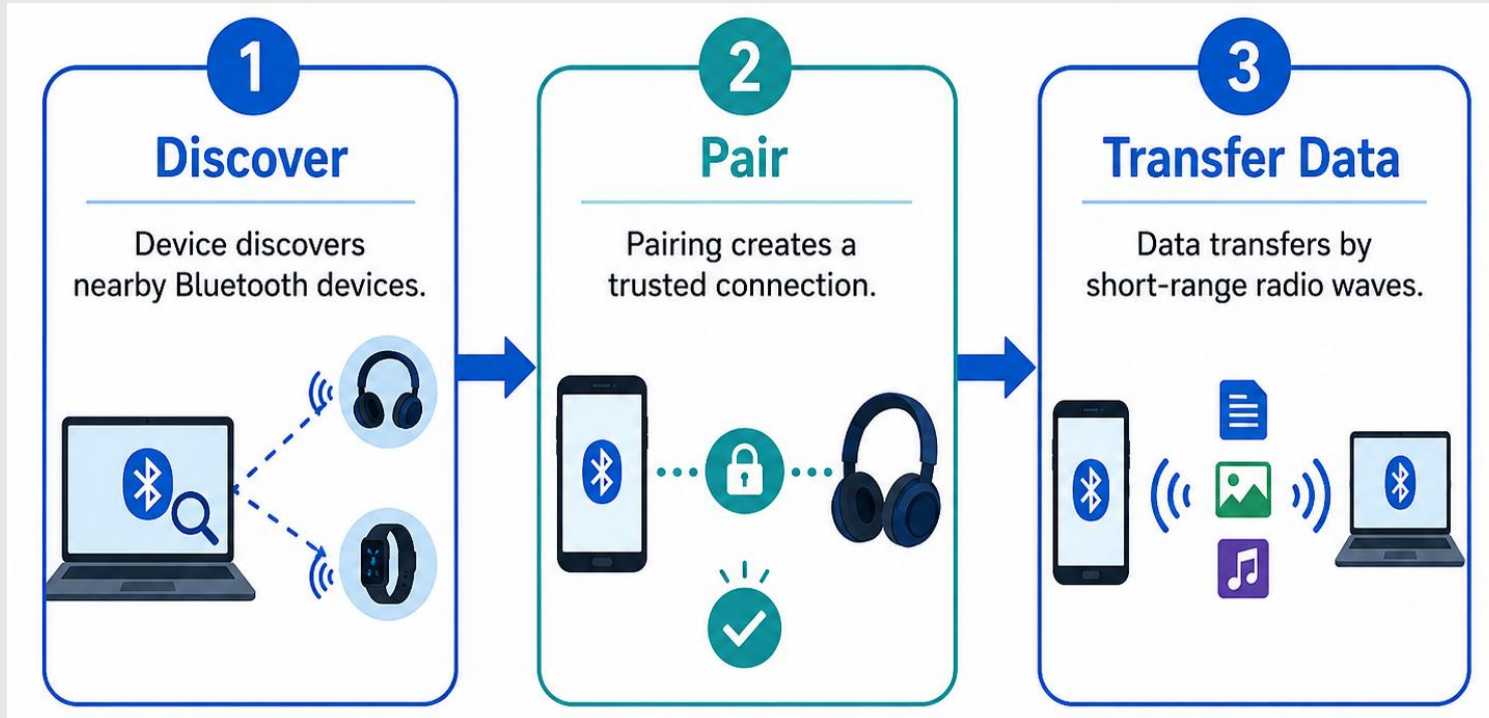


Historical Callout Box

Invented in 1994 by Ericsson as an RS-232 cable alternative. Named to honor the 10th-century Danish King "Harald Bluetooth" Gormsson, who united various tribes—mirroring the technology's goal to unite different communication protocols.



How Bluetooth Works: Pairing



Step 1: Device discovers nearby Bluetooth devices.

Step 2: Pairing creates a trusted connection.

Step 3: Data transfers by short-range radio waves.

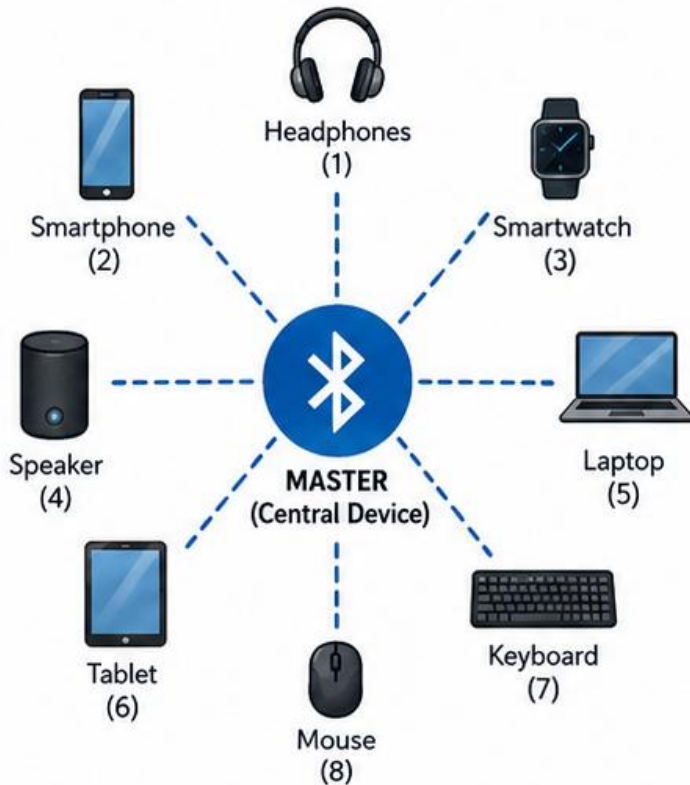


How Bluetooth Works: Piconet and Scatternet



PICONET

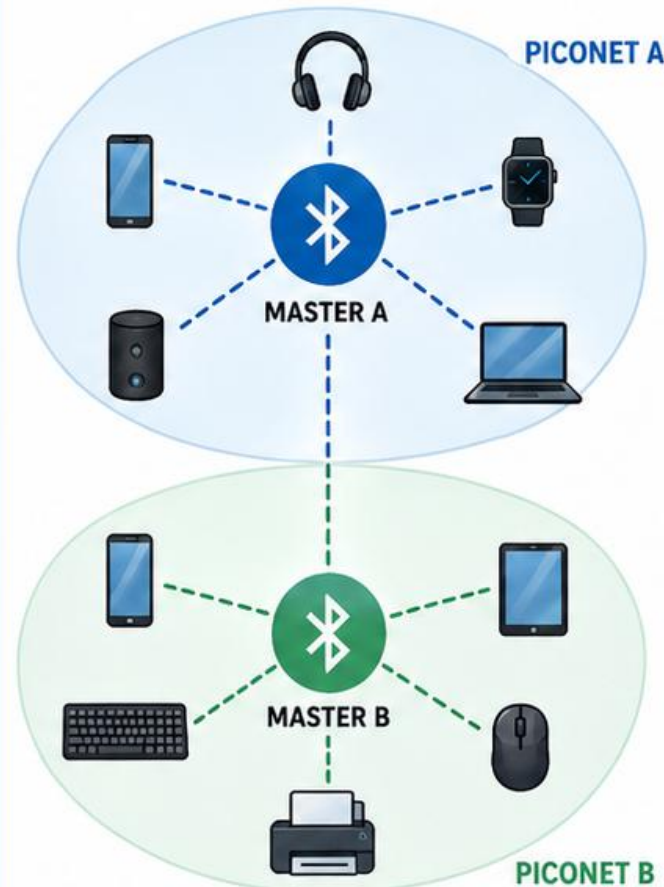
One master device can connect with up to **7** active slave devices.



Note: A piconet can have up to 7 active slave devices. More can be parked (standby).

SCATTERNET

Two or more piconets can be linked together to form a **scatternet**.



Piconet: Master device connects nearby slave devices
Multiple piconets = Scatternet

Bluetooth is like talking to a friend sitting beside you.

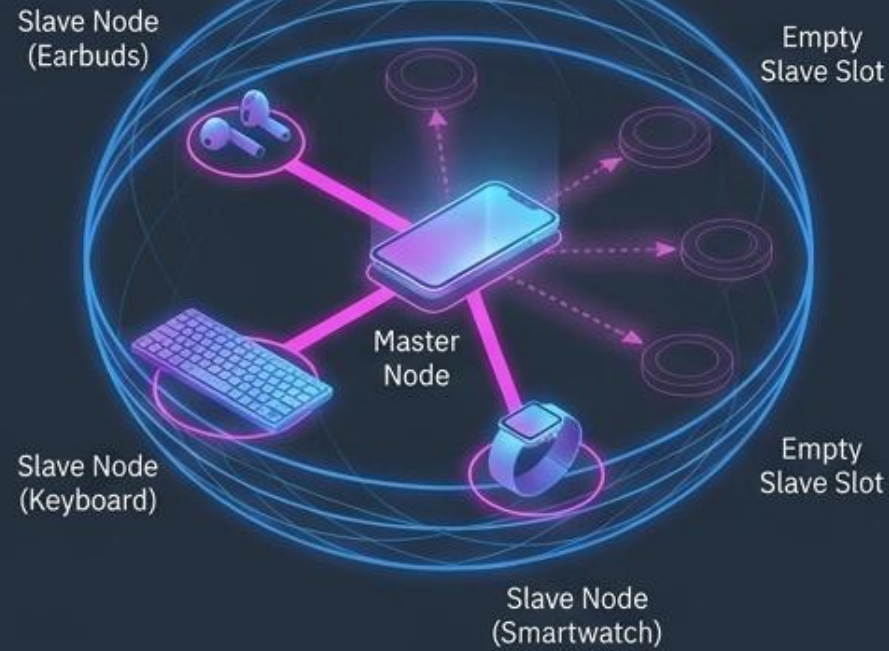
A Bluetooth small network is called a piconet.
Multiple piconets can create a scatternet.



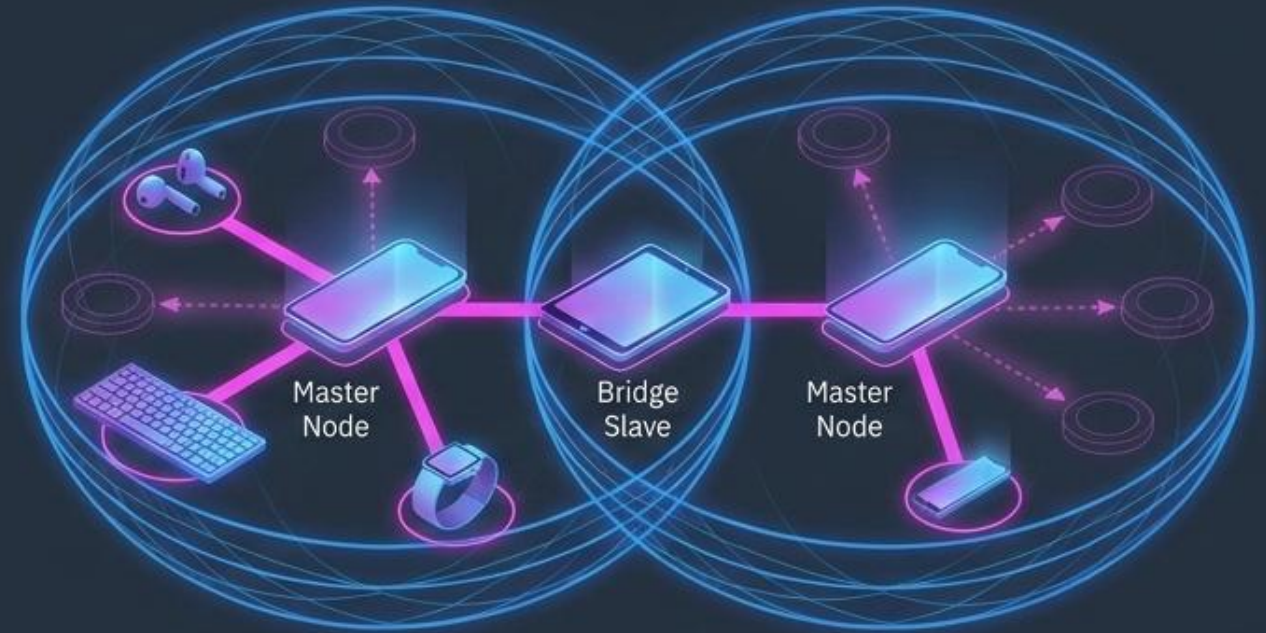
How Bluetooth Works: Piconet and Scatternet



Piconet



Scatternet





Wi-Fi: Wireless Local Area Network



Wi-Fi is used to create a Wireless Local Area Network (WLAN).

Connects phones, laptops, tablets and smart TVs to a router/access point.

Usually works in 2.4 GHz and 5 GHz bands.

Used at home, school, office, café, airport and public hotspot.

IEEE Standard: 802.11

**Network type:
WLAN**

**Range: about
50–300 m**



WAP



Internet reaches the router through cable/fiber/mobile backhaul.

Router broadcasts wireless radio signals.

Your device sends requests back to the router.

The router brings data from internet and sends it to your device.



Wi-Fi Security and Hotspot



Wi-Fi is easy to connect, so security is important.

Use strong password and encryption.

A hotspot is a fixed wireless internet coverage area.

Hotspots are common in schools, cafés, airports and libraries.

**Discussion: What makes
a Wi-Fi password weak?**

**Example: 12345678 is easy
to guess.**



Wi-Fi Security and Hotspot



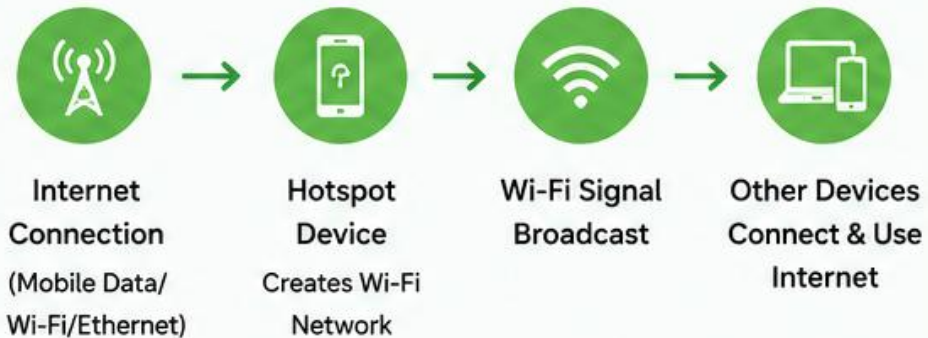
1 WHAT IS A HOTSPOT?

A hotspot is a feature that allows a device (smartphone, laptop, or dedicated device) to share its internet connection with other devices via Wi-Fi.



2 HOW DOES IT WORK?

The hotspot device uses mobile data or another internet connection and broadcasts a Wi-Fi signal. Other devices connect to this signal to access the internet.



3 TYPES OF HOTSPOT



Mobile Hotspot

Smartphone uses mobile data to share internet.



Wi-Fi Hotspot (Laptop)

Laptop connects to internet and shares via Wi-Fi.



Dedicated Hotspot Device

Portable devices made for sharing internet.



Public or Open Wi-Fi: Should we use?



1

What is it?

A free/shared Wi-Fi network anyone can join.

2



Okay for



- Browsing websites
- Watching videos
- Reading news

2



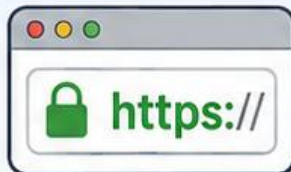
Avoid for



- Banking
- Online shopping
- Passwords & logins
- Private or sensitive data

3

Stay Safe



Use HTTPS



Use a VPN



Turn off auto-connect



Avoid file sharing



WiMAX: Broadband Over a Large Area



Full form: Worldwide Interoperability for Microwave Access.

IEEE Standard: 802.16

A high-speed wireless broadband technology.

**Coverage:
10–50 km**

Network type: WMAN / Wireless Metropolitan Area Network.

**Needs base station
+ receiver**

Useful for rural/remote areas where cable internet is difficult.



WiMAX: Broadband Over a Large Area



Technical Dashboard

Standard
IEEE 802.16 (WiMAX)

Base Coverage
10 to 50 km (up to 70 km)

Data Rate
80 Mbps up to 1+ Gbps

Transmission Mode
Full-duplex

Core Components Architecture

Base Station (Tower)



Broadcasts over massive geographical areas to thousands of clients. Operates similarly to cell towers.

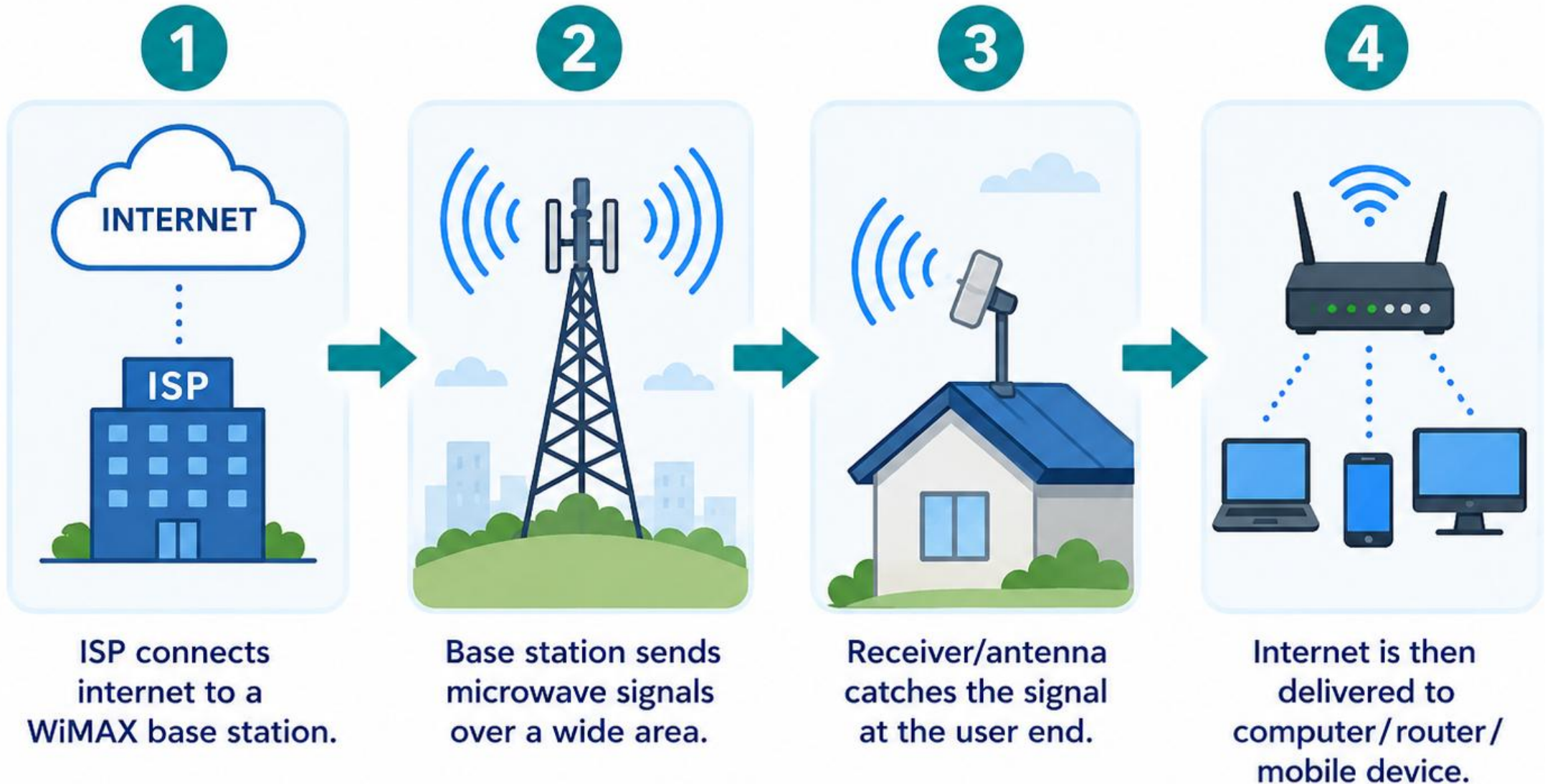
WiMAX Receiver



Built into devices or standalone modems provided by ISPs, featuring a dedicated antenna.



WiMAX: Broadband Over a Large Area



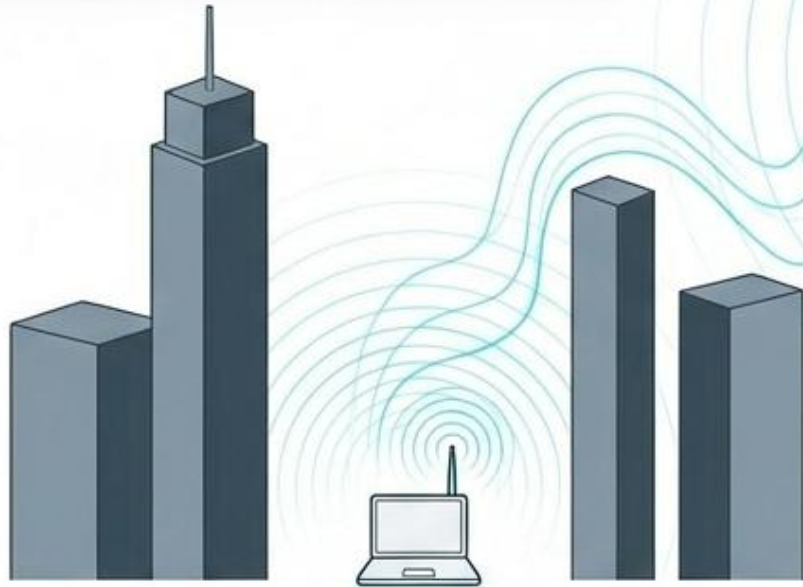


WiMAX: LOS, NLOS



Non-Line-of-Sight (NLOS)

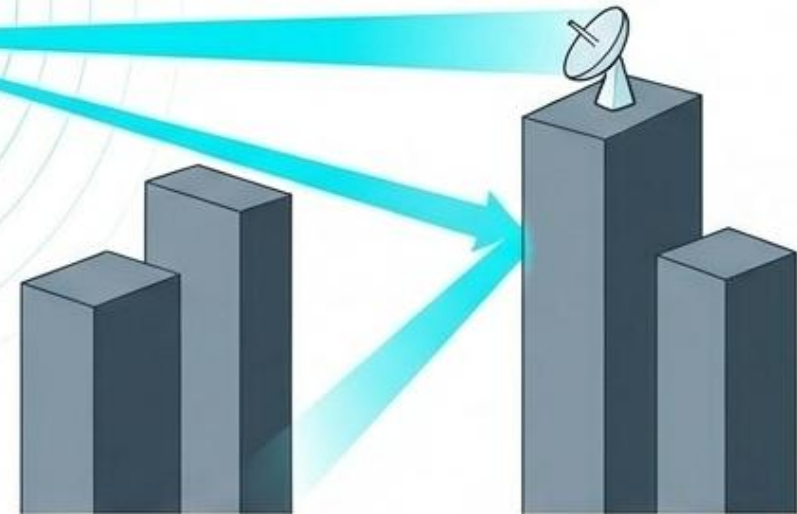
Frequency: 2 GHz to 3.5 GHz
(Lower frequency, longer wavelength).



Lower-wavelength transmissions can bypass obstacles, allowing connectivity without direct visibility.

Line-of-Sight (LOS)

Frequency: 10 GHz to 66 GHz
(Higher frequency, shorter wavelength).



Demands absolute direct visibility between tower and antenna. Rewards with vastly higher bandwidth and error-free, simultaneous data transmission.



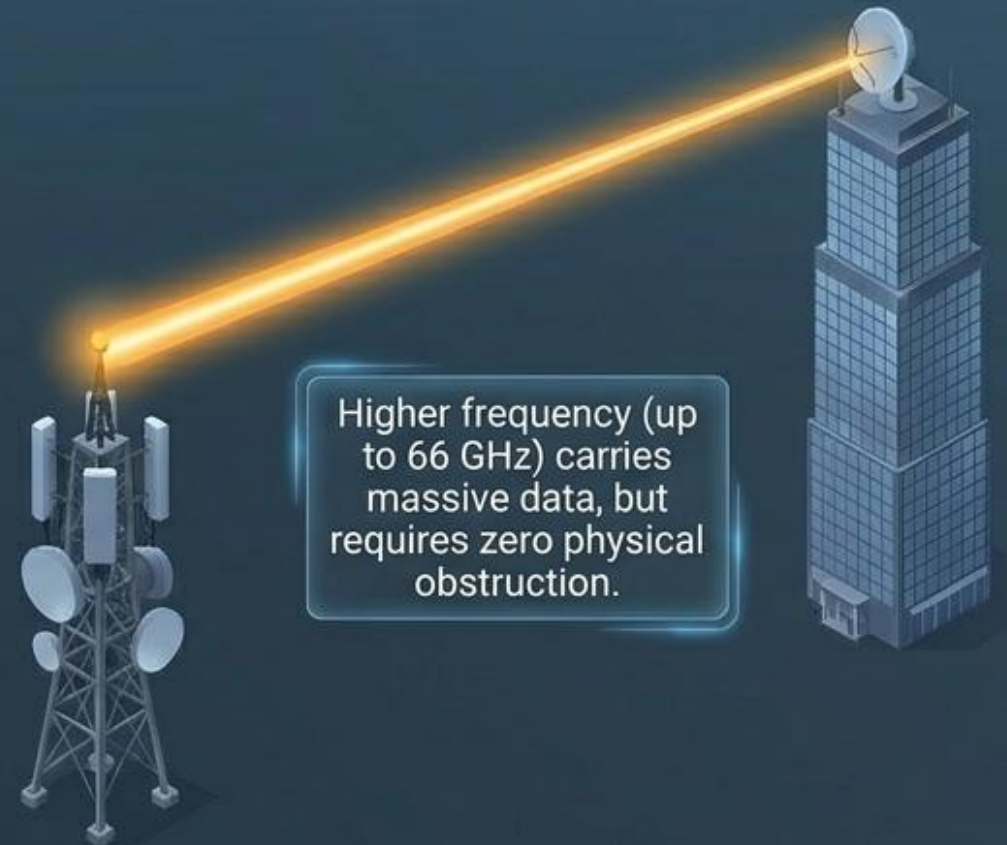
WiMAX: LOS, NLOS



Non-Line-of-Sight (NLOS)



Line-of-Sight (LOS)





WiMAX: Fixed and Mobile



Fixed WiMAX



Receiver stays at a fixed place.

- ✓ Installed at one location
- ✓ Good for home or office use
- ✓ Does not move while in use

Mobile WiMAX



Receiver can move within the service area.

- ✓ Used while moving
- ✓ Works inside the coverage area
- ✓ Suitable for portable devices



Comparison: Bluetooth vs Wi-Fi vs WiMAX



Feature	Bluetooth	Wi-Fi	WiMAX
IEEE Standard	802.15	802.11	802.16
Network Type	WPAN	WLAN	WMAN
Frequency	2.4–2.45 GHz	2.4–5 GHz	2–66 GHz
Range	3–10 m	50–300 m	10–50 km
Speed	Low	Medium–High	High
Main Use	Personal devices	Home/school internet	Wide-area broadband